

EXPERIMENT (6)

AIM

To retrieve records from Hostel Management System tables using SELECT with BETWEEN, IN and Aggregate Functions.

DESCRIPTION

BETWEEN selects values within a range. IN checks values from a list. Aggregate functions such as AVG, COUNT, SUM, MIN and MAX perform calculations on groups of rows.

SYNTAX

SELECT column_name FROM table_name WHERE column_name BETWEEN value1 AND value2;

SELECT column_name FROM table_name WHERE column_name
IN(value1,value2,...);

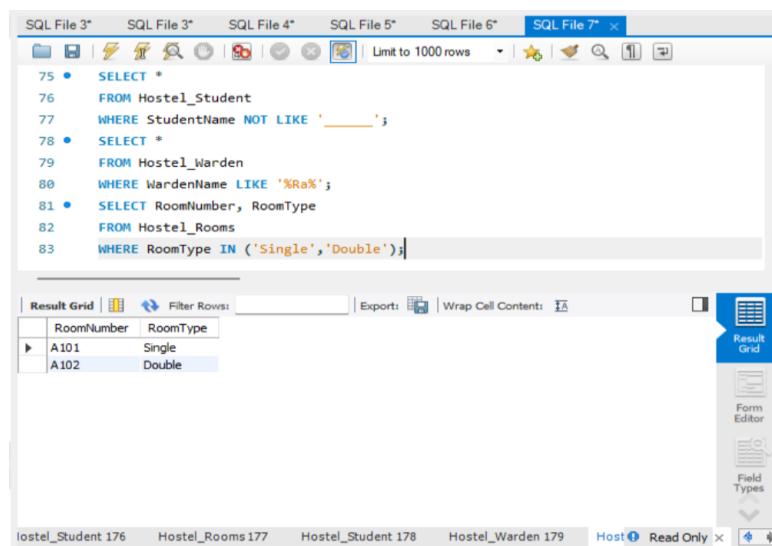
SELECT AVG(column), MIN(column), MAX(column), SUM(column),
COUNT(column) FROM table_name;

Questions and Answers

IN & BETWEEN

1. List the room numbers of rooms whose type is 'Single' or 'Double'.

SELECT RoomNumber, RoomType FROM Hostel_Rooms WHERE RoomType IN ('Single','Double');



2. Display rooms whose monthly rent is between 3500 and 5500.

SELECT * FROM Hostel_Rooms WHERE RentPerMonth BETWEEN 3500 AND 5500;

The screenshot shows a SQL IDE with a query editor and a result grid. The query editor contains the following SQL code:

```
78 • SELECT *
79 FROM Hostel_Warden
80 WHERE WardenName LIKE '%Ra%';
81 • SELECT RoomNumber, RoomType
82 FROM Hostel_Rooms
83 WHERE RoomType IN ('Single', 'Double');
84 • SELECT *
85 FROM Hostel_Rooms
86 WHERE RentPerMonth BETWEEN 3500 AND 5500;
```

The result grid displays the following data:

RoomID	RoomNumber	BlockName	FloorNo	RoomType	Capacity	OccupiedBeds	RentPerMonth	RoomStatus	WardenName
101	A101	A Block	1	Single	1	1	3500.00	Available	Mr. Ravi
102	A102	A Block	1	Double	2	1	4500.00	Available	Mr. Ravi
103	B201	B Block	2	Triple	3	2	5500.00	Occupied	Mrs. Priya

AGGREGATE

1. Find the average monthly rent of all hostel rooms.

SELECT AVG(RentPerMonth) AS Average_Rent FROM Hostel_Rooms;

The screenshot shows a SQL IDE with a query editor and a result grid. The query editor contains the following SQL code:

```
80 WHERE WardenName LIKE '%Ra%';
81 • SELECT RoomNumber, RoomType
82 FROM Hostel_Rooms
83 WHERE RoomType IN ('Single', 'Double');
84 • SELECT *
85 FROM Hostel_Rooms
86 WHERE RentPerMonth BETWEEN 3500 AND 5500;
87 • SELECT AVG(RentPerMonth) AS Average_Rent
88 FROM Hostel_Rooms;
```

The result grid displays the following data:

Average_Rent
4500.000000

2. Display the maximum and minimum monthly rent.

SELECT MAX(RentPerMonth) AS Maximum_Rent, MIN(RentPerMonth) AS Minimum_Rent FROM Hostel_Rooms;

The screenshot shows a SQL query editor with the following code:

```
83 WHERE RoomType IN ('Single', 'Double');
84 SELECT *
85 FROM Hostel_Rooms
86 WHERE RentPerMonth BETWEEN 3500 AND 5500;
87 SELECT AVG(RentPerMonth) AS Average_Rent
88 FROM Hostel_Rooms;
89 SELECT MAX(RentPerMonth) AS Maximum_Rent,
90        MIN(RentPerMonth) AS Minimum_Rent
91 FROM Hostel_Rooms;
```

The result grid below the query shows the output of the last query:

Maximum_Rent	Minimum_Rent
5500.00	3500.00

3. Display the maximum, minimum and average rent for each room type.

SELECT RoomType, MAX(RentPerMonth) AS Max_Rent, MIN(RentPerMonth) AS Min_Rent, AVG(RentPerMonth) AS Avg_Rent FROM Hostel_Rooms GROUP BY RoomType;

The screenshot shows a SQL query editor with the following code:

```
89 SELECT MAX(RentPerMonth) AS Maximum_Rent,
90        MIN(RentPerMonth) AS Minimum_Rent
91 FROM Hostel_Rooms;
92 SELECT RoomType,
93        MAX(RentPerMonth) AS Maximum_Rent,
94        MIN(RentPerMonth) AS Minimum_Rent,
95        AVG(RentPerMonth) AS Average_Rent
96 FROM Hostel_Rooms
97 GROUP BY RoomType;
```

The result grid below the query shows the output of the last query:

RoomType	Maximum_Rent	Minimum_Rent	Average_Rent
Single	3500.00	3500.00	3500.000000
Double	4500.00	4500.00	4500.000000
Triple	5500.00	5500.00	5500.000000

4. Display block name and average monthly rent.

SELECT BlockName, AVG(RentPerMonth) AS Avg_Rent FROM Hostel_Rooms
GROUP BY BlockName;

The screenshot shows a SQL IDE window titled 'SQL File 7'. The query editor contains the following SQL code:

```
93     MAX(RentPerMonth) AS Maximum_Rent,  
94     MIN(RentPerMonth) AS Minimum_Rent,  
95     AVG(RentPerMonth) AS Average_Rent  
96 FROM Hostel_Rooms  
97 GROUP BY RoomType;  
98 • SELECT BlockName,  
99     AVG(RentPerMonth) AS Average_Rent  
100 FROM Hostel_Rooms  
101 GROUP BY BlockName;
```

Below the query editor, the 'Result Grid' is displayed with the following data:

BlockName	Average_Rent
A Block	4000.000000
B Block	5500.000000

The interface includes a toolbar with icons for file operations, a 'Limit to 1000 rows' dropdown, and a sidebar with 'Result Grid', 'Form Editor', and 'Field Types' options. The status bar at the bottom shows 'Result 2: Read Only'.

5. Calculate the total monthly rent of all rooms.

SELECT SUM(RentPerMonth) AS Total_Rent FROM Hostel_Rooms;

The screenshot shows the same SQL IDE window with the following SQL code:

```
95     AVG(RentPerMonth) AS Average_Rent  
96 FROM Hostel_Rooms  
97 GROUP BY RoomType;  
98 • SELECT BlockName,  
99     AVG(RentPerMonth) AS Average_Rent  
100 FROM Hostel_Rooms  
101 GROUP BY BlockName;  
102 • SELECT SUM(RentPerMonth) AS Total_Rent  
103 FROM Hostel_Rooms;
```

The 'Result Grid' now shows the total monthly rent:

Total_Rent
13500.00

The interface elements are consistent with the previous screenshot, including the toolbar, sidebar, and status bar showing 'Result 3: Read Only'.

6. Display the number of students staying in each room.

```
SELECT RoomID, COUNT(StudentID) AS Student_Count FROM Hostel_Student  
GROUP BY RoomID;
```

The screenshot shows a SQL editor window titled 'SQL File 7'. The query is as follows:

```
99      AVG(RentPerMonth) AS Average_Rent  
100     FROM Hostel_Rooms  
101     GROUP BY BlockName;  
102 •   SELECT SUM(RentPerMonth) AS Total_Rent  
103     FROM Hostel_Rooms;  
104 •   SELECT RoomID,  
105          COUNT(StudentID) AS Student_Count  
106     FROM Hostel_Student  
107     GROUP BY RoomID;
```

The 'Result Grid' shows the following data:

RoomID	Student_Count
101	1
102	1

The bottom status bar indicates 'Result 340' is selected and 'Read Only'.

7. Display the number of rooms available in each block.

```
SELECT BlockName, COUNT(RoomID) AS Total_Rooms FROM Hostel_Rooms  
GROUP BY BlockName;
```

The screenshot shows the same SQL editor window with the following query:

```
103     FROM Hostel_Rooms;  
104 •   SELECT RoomID,  
105          COUNT(StudentID) AS Student_Count  
106     FROM Hostel_Student  
107     GROUP BY RoomID;  
108 •   SELECT BlockName,  
109          COUNT(RoomID) AS Total_Rooms  
110     FROM Hostel_Rooms  
111     GROUP BY BlockName;
```

The 'Result Grid' shows the following data:

BlockName	Total_Rooms
A Block	2
B Block	1

The bottom status bar indicates 'Result 368' is selected and 'Read Only'.

RESULT

Thus, the records were successfully retrieved using SELECT statements with BETWEEN, IN and Aggregate Functions on the Hostel Management System tables.